Basic Info.

Model NO.	KS-X5000P
Test Types	Flaky Material
Brand	Kathmatic
Detection Method	Optics
Automatic Grade	Fully Automatic
Type	Single-point
Application Area	Quality Control
Pipeline System	Integrated
Z-Axis Drive Travel	50mm
Minimum Step Accuracy	0.1μm
Image Sensor	25 Million High-Definition Pixels
Main Control Processor	Intel I7 High-Performance Processor
Operating Memory	64GB Large Memory

Data Storage	1tb SSD High-Speed Solid-State Drive
Independent Graphics Card	Rtx3060 Professional Graphics Card
Display Device	27-Inch 4K Ultra-Clear Screen,Resolution 3840×2160
Operating Temperature	5℃-40℃
Operating Humidity	5%-80% Rh,No Condensation
Power Supply System	Equipped with Industrial-Grade Stable Power Supply
Transport Package	Cardboard Box
Specification	Customized
Trademark	Kathmatic
Origin	China

Packaging & Delivery

Package Size	75.00cm * 45.00cm * 62.00cm
Package Gross Weight	112.000kg
Lead Time	30 days (1 - 1 Sets) To be negotiated (> 1 Sets)

Product Description

Main machine			
Model	KS-X5000P	KS-X5050P	KS-X5000A
Focus Z-axis drive mode	Motorized stage		
Focus Z-axis movable range(mm)	50	100	50
Stage Z-axis drive mode	Motorized	/	Motorized
Stage Z-axis movable range(mm)	50	/	50
Sensor pixels	12 million		5 million
Objective lens switch mode	Motorized		Manual
Magnification switch mode	Motorized		
CPU	Intel i7		
Memory	64GB		
Storage	1TB SSD		
Graphics card	RTX3060		
Display resolution	3840x2160		
Display size	27-inch		
Operating environment temperature	5~40℃		
Operating environment humidity	5%-80% RH, no condensation		

XY stage				
Stage model	KS-MS100a	KS-MS050R	KS-MS200	KS-MS310
Size(mm)	284x235	203x187	426x315	576x524
Movable range(mm)	100x100	50x50	200x160	310x310
Rotation angle	/	±90°	/	/
Drive mode	Motorized			
Load capacity	5kg			30kg

Objective lens(compatible with KS-X5000P/KS-X5050P)				
Model	KS-LM111	KS-LM112	KS-LM113	KS-LM114
Magnification	20	45	100	200
	30	67.5	150	300
	40	90	200	400
	60	135	300	600
	100	225	500	1000
	120	270	600	1200
Support for objective lens model identification				

Product Description

Kathmatic Ultra DOF 3D Digital Microscope focuses on the core technologies of ultra depth synthesis, ultra high definition image stitching, multi-mode intelligent lighting, fully electric high-precision measurement, and AI intelligent image optimization. It comprehensively covers fields such as

semiconductor, lithium battery, photovoltaic, 3C electronics, automotive manufacturing, aerospace, new material research and development, precision processing, cultural and archaeological research, and biological research, providing a one-stop non-contact non-destructive solution for microscopic observation, 3D morphology analysis, precision dimension measurement, defect detection, and failure analysis.

KS Series

3D inspection with large DOF, wide FOV and real-color

Ultra-depth 3D imaging

Ultra-high resolution and large-format imaging

By utilizing deep synthesis technology, global focusing is achieved, generating a true-color 3D point cloud model that can be used for stereoscopic observation and quantitative analysis of sample surface topography, elevation variations, and microstructure.

With an optical resolution of 250nm and a 25-million-pixel sensor, it supports 100,000 × 100,000 pixel stitching, enabling high-precision imaging and archiving of large-sized samples.

Multispectral and Polarized Illumination System

High-precision measurement and analysis

Intelligent scientific research tools

Equipped with four-color light sources, polarization, and differential interference illumination, it can enhance structural contrast, eliminate reflection, identify stress distribution, and improve the clarity of transparent sample edges, meeting the observation needs of various scientific research samples.

Supports 2D/3D geometric measurement, roughness measurement, volume/area statistics, particle size analysis, and contour analysis. The data can be exported, repeated, and compared, meeting the requirements of scientific research papers and project experimental data.

With functions such as split-screen comparison, image navigation, real-time focusing, playback shooting, image stabilization, electron microscope simulation, scheduled shooting, and time-lapse photography, it fully meets the needs of scientific observation and experimental recording.